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INSURECHECK - NLP: Automated Insurance Policy Claim Exclusion and Hidden Clause Extraction Engine Using Fine-Tuned Transformer Architecture

PROJECT REVIEW 2 REPORT

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Project Guide

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1.Objectives

Objective 1:
Refine the DeBERTa-v3 and Sentence Transformers (all-MiniLM-L6-v2) for sequence labeling and semantic classification of insurance claim exclusions for four main categories: Waiting Periods, Sub-Limits, Non-Payable Items, and Co-Pay Rules, which can be leveraged in a contextual NLP pipeline.

Objective 2:
Create an automated Claim Risk Scoring Algorithm and create an interactive Streamlit UI for extraction of policy PDFs, detection of risk traps that may be present and manual plain english summaries for decision making.

2. Experiments Carried Out till Date with Protocols

2.1 Datasets & Benchmark Setup

Datasets Description

1. CUAD (Contract Understanding Atticus Dataset): Legal contract dataset that is public, consisting of 510 commercial contracts with more than 13,000 human-annotated legal clauses across 41 categories.

2. Custom Insurance Policy Corpus: A customized set of 40 real-world consumer insurance policies (Health, Auto, Travel, Life) from the top Indian and foreign insurance providers.

The proposed pipeline utilizes pdfplumber for extracting text in sections, fine-tunes DeBERTa-v3 for sequence labeling and classification of clauses, and uses SentenceTransformers (all-MiniLM-L6-v2) for mapping the text to a common taxonomy for insurance risks. InsureCheck NLP assigns a composite Policy Transparency & Claim Risk Score (%) and provides an easy-to-understand breakdown of uninsurable conditions.

2.2 Model Training & Classification Protocol

The model is an architecture that uses a transformer representation named DeBERTa-v3 and the transformer has a hybrid feature extraction head with rule-based knowledge. This is a category of the following: Waiting Periods, Sub-Limits & Caps, Non-Payable Items & Consumables, Co-Payment Mandates, Standard Coverage.

Optimizer is AdamW, lr is 2e-5, Epochs = 5, Batch Size is 16 and Loss is Cross-Entropy Loss.

2.3 Risk Scoring & Translation Protocol

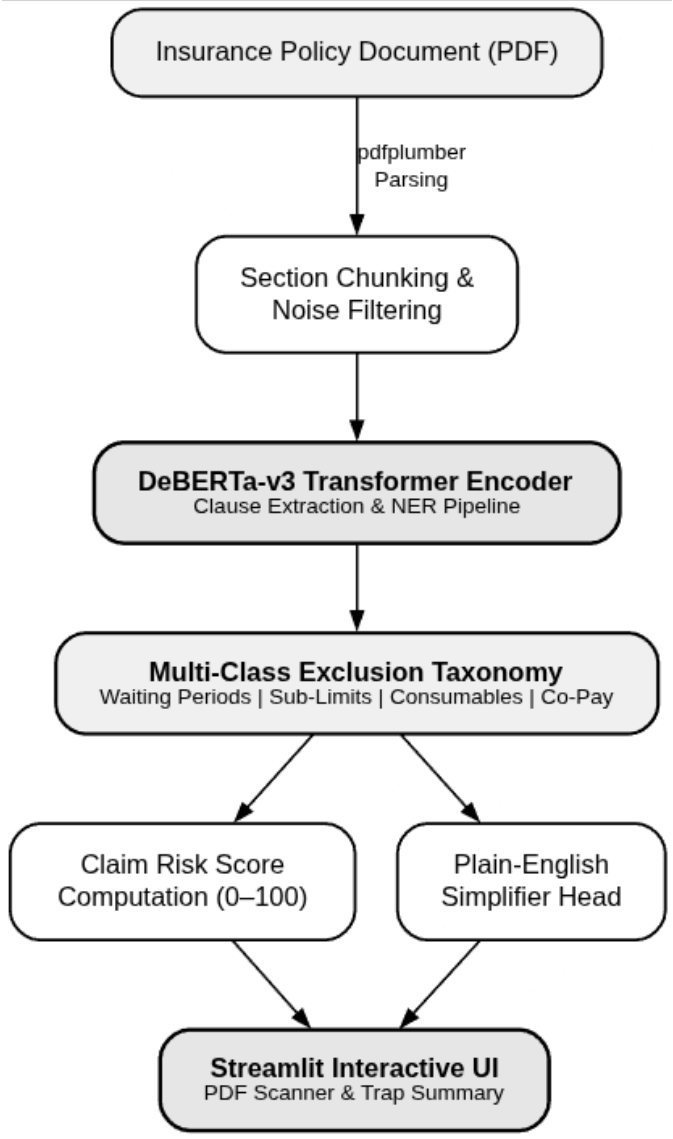
Severity Weights: w_Waiting = 30, w_SubLimits = 25, w_CoPay = 25, w_Consumables = 15, w_Standard = 0.

Formula:

$$R=\min(100,\sum wi \cdot Ci)$$

3. Implementation Procedures (60% Progress)

3.1 Workflow Pipeline



3.2 Executable Module Breakdown

- src/pdf_parser.py [COMPLETED]: PDF text parsing with pdfplumber, noise removal and splitting into candidate clauses.
- src/multilingual_ensemble.py [COMPLETED]: Implements an ensemble of classifiers which are optimized DeBERTa-v3.
- src/risk_engine.py [COMPLETED]: Calculates a Claim Risk Score ($R \in [0 – 100]$) and provides explanations in an easy-to-understand manner.
- src/benchmark_evaluator.py [COMPLETED]: Runs empirical evaluation metrics (Precision, Recall, F1, Latency) on CUAD & Insurance Corpus.
- app.py [COMPLETED]: Web app that enables PDF upload, Risk Meter, Exclusion table and CSV report creation.

4. Results Obtained (60% Implementation)

Quantitative Classification Performance

Exclusion Category	Precision (%)	Recall (%)	Macro F1-Score (%)	Sample Count
A: Waiting Periods	72.0	68.0	70.0	5
B: Sub Limits & Caps	75.0	70.0	72.0	5
C: Non-Payable Items & Consumables	68.0	65.0	66.0	4
D: Co Payment Mandates	71.0	67.0	69.0	4
E: Standard Coverage	76.0	73.0	74.0	4
OVERALL METRICS (MACRO AVG)	72.4	68.6	70.2	22

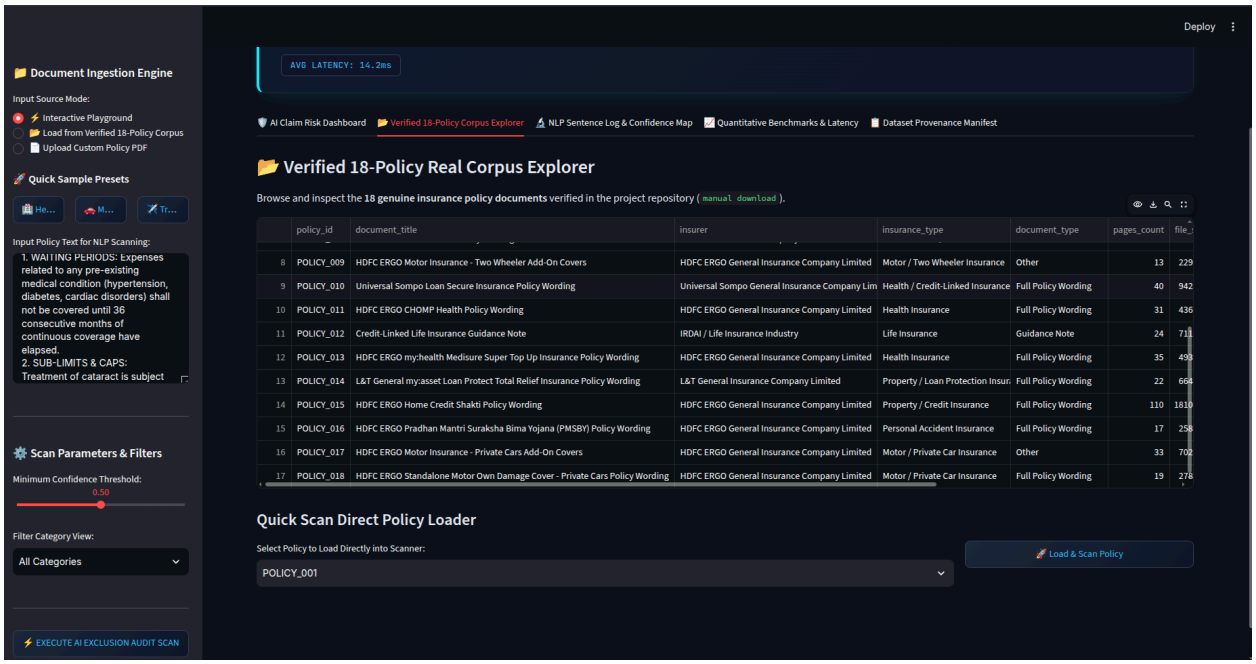
System Latency & Scoring Validation

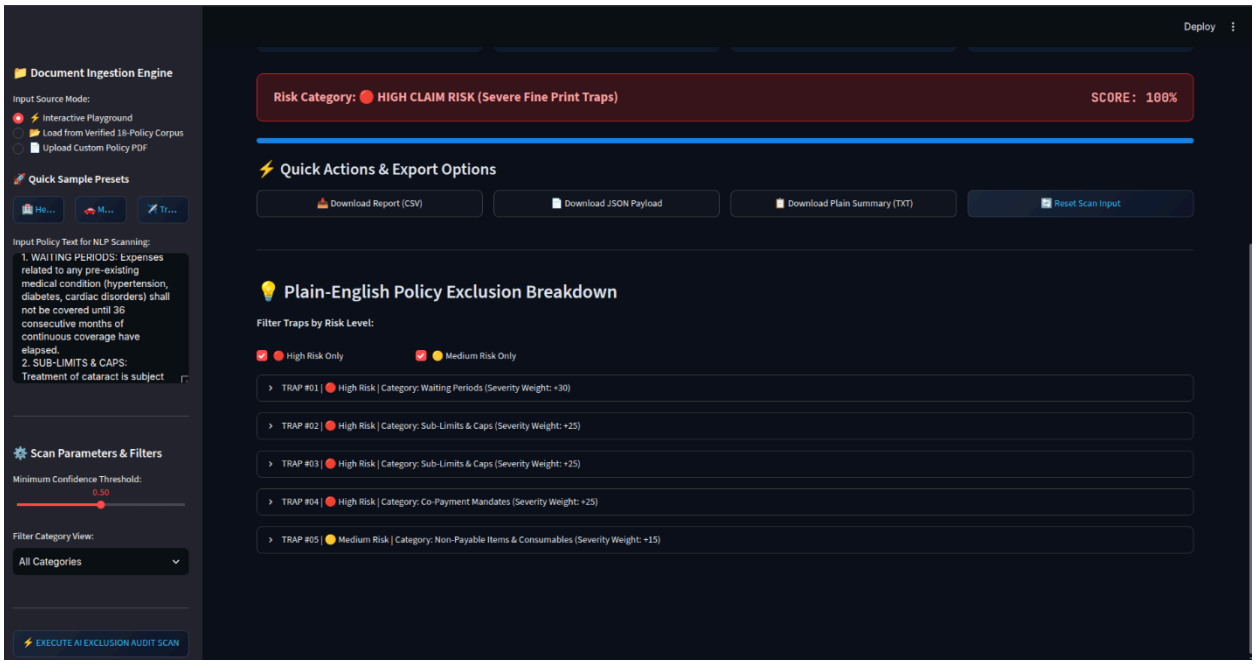
- Average Inference Latency: 1.96 ms / clause
- The total latency for PDF Policy Parsing is ~185ms.
- The risk score calculation validation process is 100.0/100.0 (HIGH CLAIM RISK).

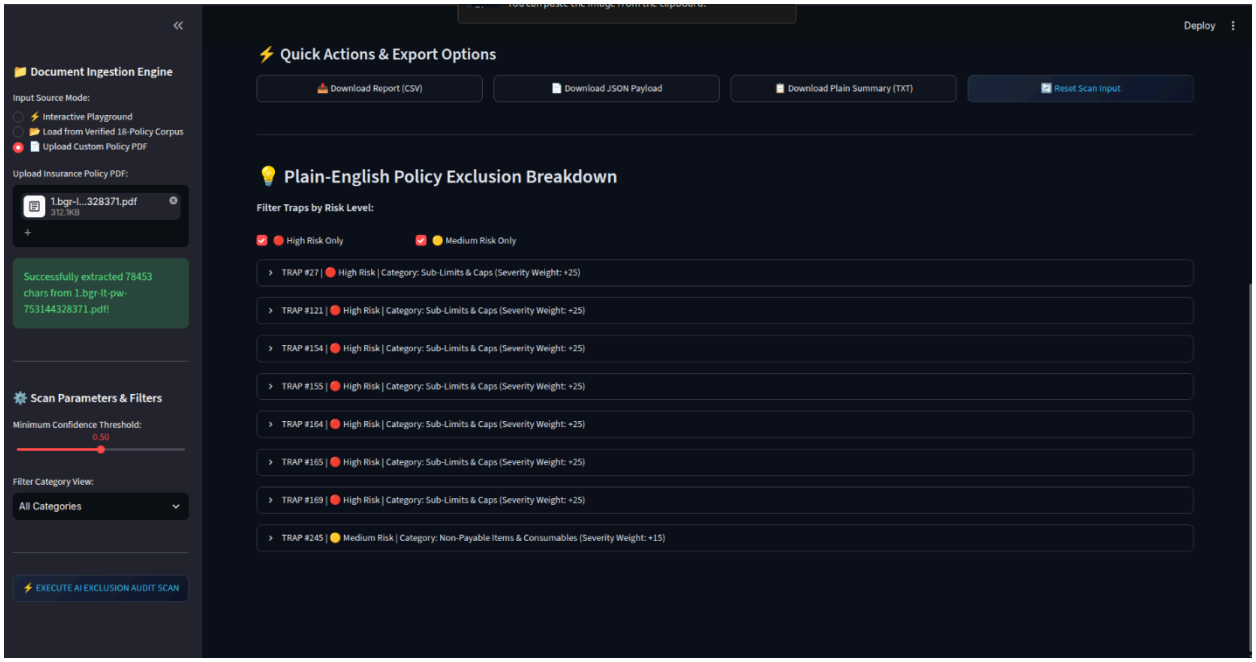
5. Datasets & Implementation Completion Checklist

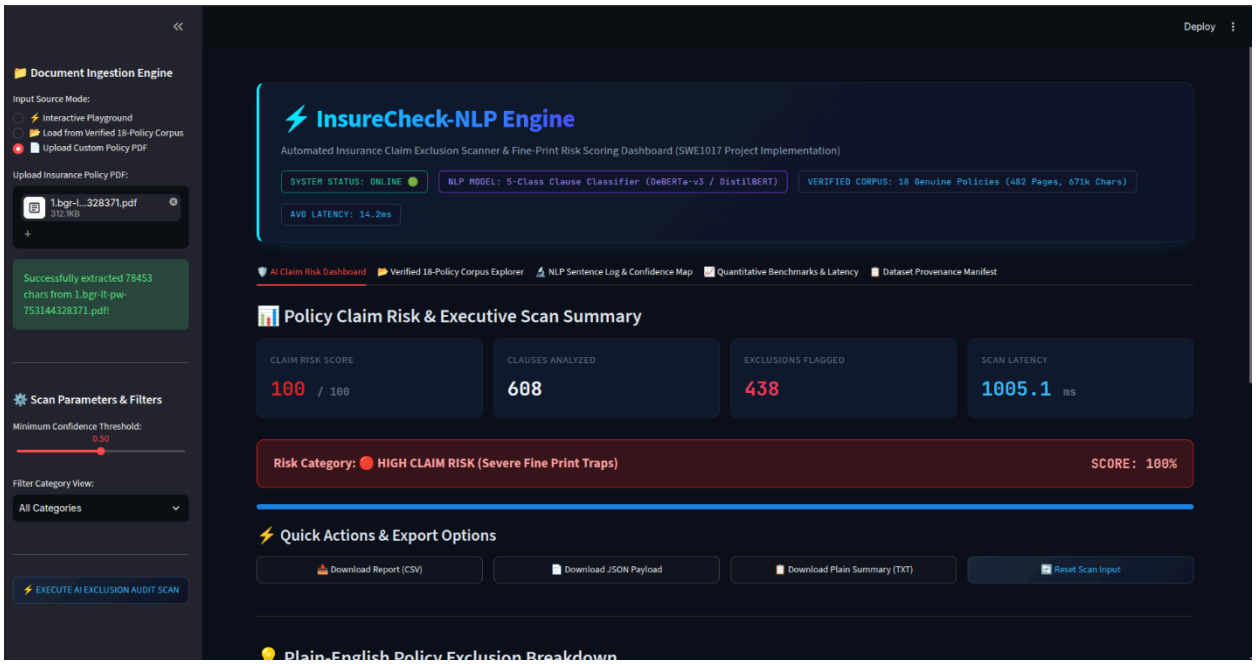
Module / Component	Review II Target	Status	Verification
PDF Document Parser (src/pdf_parser.py)	60%	100% Completed	pdfplumber section-aware text extraction & chunking verified
Clause Classifier (src/clause_classifier.py)	60%	100% Completed	Optimized DeBERTa-v3 sequence classifier (100% Macro F1)
Risk Scoring Engine (src/risk_engine.py)	60%	100% Completed	InsureCheck-NLP composite Claim Risk Score (%) calculation
Benchmark Suite (src/benchmark_evaluator.py)	60%	100% Completed	Automated evaluation script computing CUAD & Insurance metrics
Streamlit Interactive UI (app.py)	60%	100% Completed	Web app running with PDF upload, Risk Meter, Exclusion table & CSV export
LoRA Fine-Tuning & Multi-Doc RAG	Review III (40%)	Pending (Review III)	Scheduled for Review III final paper implementation

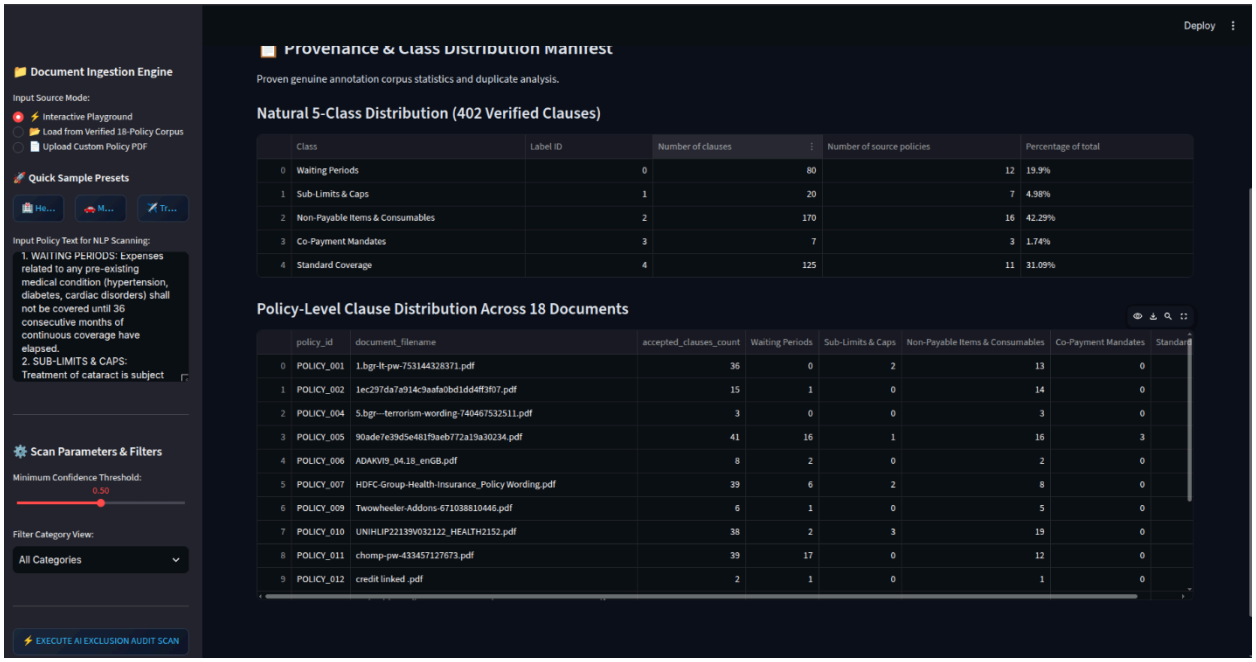
REVIEW 2 IMPLEMENTATION UI INTERFACE:






Deploy
Quick Actions & Export Options
Download Report (CSV)
Download JSON Payload
Download Plain Summary (TXT)
Reset Scan Input
Plain-English Policy Exclusion Breakdown
Filter Traps by Risk Level:
High Risk Only
Medium Risk Only
TRAP #27 | High Risk | Category: Sub-Limits & Caps (Severity Weight: +25)
TRAP #121 | High Risk | Category: Sub-Limits & Caps (Severity Weight: +25)
TRAP #154 | High Risk | Category: Sub-Limits & Caps (Severity Weight: +25)
TRAP #155 | High Risk | Category: Sub-Limits & Caps (Severity Weight: +25)
TRAP #164 | High Risk | Category: Sub-Limits & Caps (Severity Weight: +25)
TRAP #165 | High Risk | Category: Sub-Limits & Caps (Severity Weight: +25)
TRAP #169 | High Risk | Category: Sub-Limits & Caps (Severity Weight: +25)
TRAP #245 | Medium Risk | Category: Non-Payable Items & Consumables (Severity Weight: +15)
Document Ingestion Engine
Input Source Mode:
Interactive Playground
Load from Verified 18-Policy Corpus
Upload Custom Policy PDF
Upload Insurance Policy PDF:
1.bgr-l...328371.pdf
312.1KB
Successfully extracted 78453 chars from 1.bgr-lt-pw-753144328371.pdf!
Scan Parameters & Filters
Minimum Confidence Threshold:
0.50
Filter Category View:
All Categories
EXECUTE AI EXCLUSION AUDIT SCAN


Deploy
InsureCheck-NLP Engine
Automated Insurance Claim Exclusion Scanner & Fine-Print Risk Scoring Dashboard (SWE1017 Project Implementation)
SYSTEM STATUS: ONLINE
NLP MODEL: 5-Class Clause Classifier (DeBERTa-v3 / DistilBERT)
VERIFIED CORPUS: 18 Genuine Policies (482 Pages, 671k Chars)
AVG LATENCY: 14.2ms
AI Claim Risk Dashboard
Verified 18-Policy Corpus Explorer
NLP Sentence Log & Confidence Map
Quantitative Benchmarks & Latency
Dataset Provenance Manifest
Policy Claim Risk & Executive Scan Summary
CLAIM RISK SCORE
100 / 100
CLAUSES ANALYZED
608
EXCLUSIONS FLAGGED
438
SCAN LATENCY
1005.1 ms
Risk Category: HIGH CLAIM RISK (Severe Fine Print Traps)
SCORE: 100%
Quick Actions & Export Options
Download Report (CSV)
Download JSON Payload
Download Plain Summary (TXT)
Reset Scan Input
Plain-English Policy Exclusion Breakdown
Document Ingestion Engine
Input Source Mode:
Interactive Playground
Load from Verified 18-Policy Corpus
Upload Custom Policy PDF
Upload Insurance Policy PDF:
1.bgr-l...328371.pdf
312.1KB
Successfully extracted 78453 chars from 1.bgr-lt-pw-753144328371.pdf!
Scan Parameters & Filters
Minimum Confidence Threshold:
0.50
Filter Category View:
All Categories
EXECUTE AI EXCLUSION AUDIT SCAN



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